

CoRoT-2b

R-1641

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_210722 · created 2026-08-02T06:55:32+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459417.7199 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.002 +/- 0.044
Transit depth (Rp/Rs)^2	0.00046 +/- 0.019 [%]
Orbital Inclination (inc)	87.6 +/- 2.08
Airmass coefficient 1 (a1)	1.311 +/- 0.1
Airmass coefficient 2 (a2)	-0.204 +/- 0.077
Scatter in the residuals of the lightcurve fit is	7.47 %
Best Comparison Star	#1 - [569, 264]
Optimal Aperture	3.38
Optimal Annulus	19.68
Transit Duration (day)	0.0799 +/- 0.0082

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.002 ± 0.044 vs 0.1667 (deviation 3.49σ)
Duration fit vs published	0.0799 d vs 0.095 d (deviation 1.72σ)
Mid-time O–C	-5.3 min
Depth SNR	0.0
Residual scatter	7.47 %

Light curve

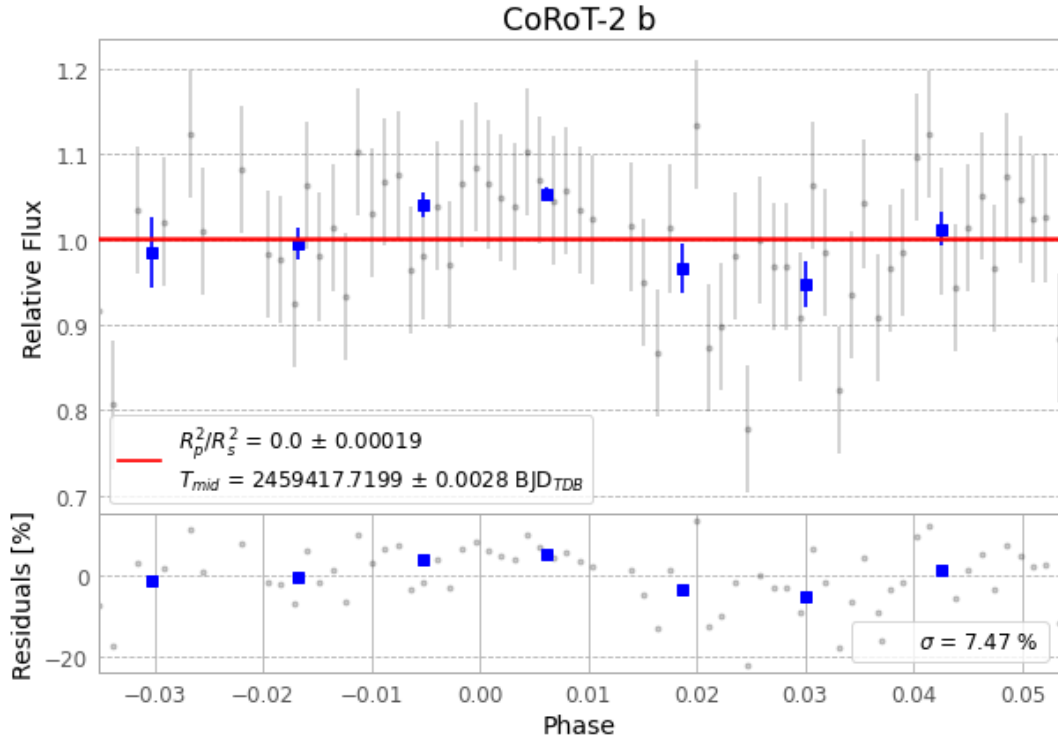


Figure 1 — FinalLightCurve_CoRoT-2 b_2021-07-21.png (SHA-256 2c56674046f1ce4b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 116], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fa23aa7289865468...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), CoRoT-2210722033912.FITS → CoRoT-2210722072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-210722.log (de281a1065ce...), FinalLightCurve_CoRoT-2 b_2021-07-21.png (2c56674046f1...), FinalLightCurve_CoRoT-2 b_2021-07-21.csv (e01774c7e581...)

Compute resources

Wall time 180.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 06:55Z.